

JAGANATH Solutions Pvt. Ltd.

Payment Gateway Management & Transaction Analytics System

Business Analyst Portfolio Project

Portfolio Demonstration — Fictional Case Study

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“Executive Summary: This project analyzes payment reliability using a controlled synthetic dataset of 5,000 transactions covering January–June 2026. The analysis identifies key failure patterns across failure reasons, payment methods, banking partners, merchants and processing time. The three leading failure reasons account for approximately 50.3% of observed failures, while Net Banking records the highest payment-method failure rate at 10.82%. Based on these findings, targeted recommendations are proposed to support payment reliability and data-driven decision-making.”

1. Company Overview + Business Problem

Company Overview

JAGANATH Solutions Pvt. Ltd. is a fictional FinTech company providing payment gateway services to online merchants. The platform supports UPI, credit/debit cards, net banking, and digital wallets. The payment journey connects customers, merchants, the JAGANATH gateway, banks/payment networks, authentication services, settlement, and refund/reconciliation activities.

Business Scenario

During January–June 2026, the payment transaction dataset showed variation in payment success performance, with the overall success rate declining from 94.59% in January to 92.52% in June. The corresponding failure rate increased from 5.41% to 7.48%. This trend provides the starting point for investigating the major failure reasons, payment methods, banking partners, merchants and operational factors associated with unsuccessful transactions.

Business Problem Statement

The analysis indicates an increase in unsuccessful payment transactions during the January–June 2026 period, while the business lacks sufficient analytical visibility into where and why failures occur. Failures may be distributed across payment methods, banking partners, merchants, transaction periods, geographical regions and other dimensions.

- Revenue leakage from unsuccessful payment attempts.

- Customer payment drop-offs and repeated payment retries.
- Lower merchant conversion and satisfaction.
- Higher customer-support queries and operational workload.
- Increased refund, reversal, and reconciliation activity.
- Difficulty identifying the highest-impact failure drivers.

Business Goal

Analyze the payment transaction lifecycle, identify the primary drivers of payment failures, quantify their business impact, establish practical KPIs, and recommend improvements that can increase payment reliability and support better decision-making.

2. Business Requirements Document (BRD)

Business Objective

Identify major payment failure drivers and provide data-driven recommendations to improve transaction success, customer experience, merchant satisfaction, and business performance.

Stakeholders

Stakeholder	Primary Need
Product Manager	Business priorities and improvement decisions
Operations	Failure patterns and operational visibility
Business Analyst	Requirements, analysis, process and recommendations
Engineering	Technical feasibility and implementation support
Finance	Revenue and business-impact view
Customer Support	Complaint and customer-experience visibility
QA	Validation requirements

In Scope

- Transactions and payment methods
- Banks, merchants and customers
- Refunds, support and processing time
- Excel/Power BI analysis
- Requirements, process analysis, stakeholder analysis, KPIs and recommendations

Out of Scope

- Production banking infrastructure
- Real financial transactions
- Live APIs
- Production deployment

- Real customer PII

Key Business Requirements

- BR-01: Provide visibility into total transaction volume and payment outcomes.
- BR-02: Measure success and failure rates.
- BR-03: Identify top payment failure reasons.
- BR-04: Identify payment methods with high failure rates.
- BR-05: Enable analysis by relevant dimensions such as bank, merchant, region and time period.
- BR-06: Provide actionable recommendations linked to analytical findings.

Assumptions

- The synthetic dataset is assumed to represent realistic payment-gateway transaction patterns for analytical demonstration.
- Bank and merchant names are illustrative and used for analytical purposes only.
- Transaction outcomes and patterns in the dataset are assumed to be suitable for demonstrating Business Analysis techniques.

Constraints

- No access to live banking APIs or production systems.
- Analysis is limited to the January–June 2026 synthetic transaction dataset.
- No real customer financial information or PII is used.
- Business impact is interpreted as expected impact where actual production loss data is unavailable.

3. Functional Requirements Document (FRD)

The functional requirements below translate the business needs into a simple analytics solution suitable for an entry-level BA portfolio.

FR ID	Functional Requirement	Expected Output
FR-01	Load and organize transaction data	Structured transaction dataset
FR-02	Classify payment outcome	Success / Failure status
FR-03	Calculate transaction KPIs	Total transactions, success %, failure %
FR-04	Analyze failure reasons	Ranked failure-reason distribution
FR-05	Analyze payment methods	Method-wise transaction and failure view
FR-06	Analyze business dimensions	Bank / merchant / region / period views
FR-07	Create dashboard	Management-friendly Power BI dashboard
FR-08	Highlight exceptions	High-failure methods, partners or periods
FR-09	Support recommendations	Insights traceable to data findings
FR-10	Validate outputs	KPI and filter results checked against source data

Non-Functional Expectations

- Simple and easy-to-understand dashboard layout.
- Consistent KPI definitions across analysis and reporting.
- Synthetic data only; no real customer financial information.
- Filters should provide clear and relevant business views.
- Results should be understandable to non-technical stakeholders.

Acceptance Criteria

- All core KPIs reconcile with the transaction dataset.
- Failure reasons can be ranked from highest to lowest.
- Payment methods can be compared on failure performance.
- Dashboard communicates the major problem areas without requiring technical knowledge.

4. AS-IS & TO-BE Process

AS-IS Payment Process

The current payment flow starts when a customer initiates payment on a merchant website/application. The request is sent to the JAGANATH gateway, validated and routed to the relevant bank or payment network. Authentication and processing occur, after which a response is returned. The transaction is then marked successful, failed, or requiring further processing. Successful transactions proceed to settlement; failed/reversed transactions may enter retry, refund or reconciliation activities.

Step	AS-IS Flow
1	Customer initiates payment
2	Merchant sends request to JAGANATH Gateway
3	Gateway validates transaction
4	Request routed to bank / payment network
5	Authentication and processing
6	Success / Failure / Further processing response
7	Merchant + customer receive status
8	Settlement / refund / reconciliation where applicable

Observed / Potential Pain Points

- Increasing payment transaction failures.
- Repeated customer retries.
- Merchant dissatisfaction and support queries.
- Potential revenue leakage.
- Higher refund/reconciliation workload.
- Dependency on multiple banking and payment-network partners.

- Limited visibility into failure distribution and root causes.

TO-BE Process

Customer → Merchant → Gateway Validation → Intelligent Failure Monitoring → Bank/Payment Network → Response → Outcome Classification → Customer/Merchant Notification → Settlement / Controlled Retry / Refund-Reconciliation.

Improvement Concept

The TO-BE process adds stronger monitoring and analytical visibility around the transaction outcome so that recurring failure patterns can be identified, prioritized and addressed.

5. Dataset Overview + Analysis Summary

Dataset Approach

The project assumes a controlled synthetic transaction dataset created specifically to represent realistic payment-gateway behavior. The source document states that the analysis is not based on real financial transactions or real customer PII.

Analytical Fields

Field	Purpose
Transaction ID	Unique transaction reference
Transaction Date/Time	Trend and period analysis
Payment Method	UPI / Card / Net Banking / Wallet comparison
Bank / Payment Partner	Partner-level performance analysis
Merchant	Merchant-level performance analysis
Region	Geographical comparison
Transaction Status	Success / Failure outcome
Failure Reason	Root-cause categorization
Amount	Transaction-value and impact analysis
Processing Time	Operational performance analysis
Refund / Reversal Flag	Post-transaction analysis

Core KPI Definitions

KPI	Definition
Total Transactions	Count of all transaction records
Success Rate	$\text{Successful transactions} \div \text{total transactions} \times 100$
Failure Rate	$\text{Failed transactions} \div \text{total transactions} \times 100$

Top Failure Reason	Failure category with the highest failed-transaction count
Top Failed Payment Method	Payment method with the highest failed-transaction count/rate

Analysis Summary

The primary analytical objective is to move from an overall success-rate problem to a focused view of where failures occur, why they occur, and which areas should be prioritized. The source document specifically proposes analysis across payment methods, banks, merchants, regions and transaction periods.

Final analytical dataset: The analysis uses 5,000 synthetic payment transactions covering January to June 2026. The dataset was validated in Excel and analyzed in Power BI using transaction outcomes, failure reasons, payment methods, banking partners, merchants, refunds and processing time.

6. Power BI Dashboard Explanation

Dashboard Purpose

The Power BI report provides a two-page management and operational view of payment performance. Page 1 focuses on executive payment performance and failure patterns, while Page 2 provides merchant and operational drill-down analysis.

Recommended Dashboard Layout

Section	Visual	Business Question
KPI Cards	Total Transactions Success Rate Failure Rate	How is overall payment performance?
Failure Analysis	Bar chart: Failure Reason	Why are payments failing?
Payment Method	Column/bar chart: Method vs Failure Rate	Which method needs attention?
Trend	Line chart: Success/Failure over time	When is performance deteriorating?
Partner View	Table/bar chart: Bank or Partner	Which partners show high failure levels?
Filters	Date Method Bank Merchant Region	Where is the issue concentrated?

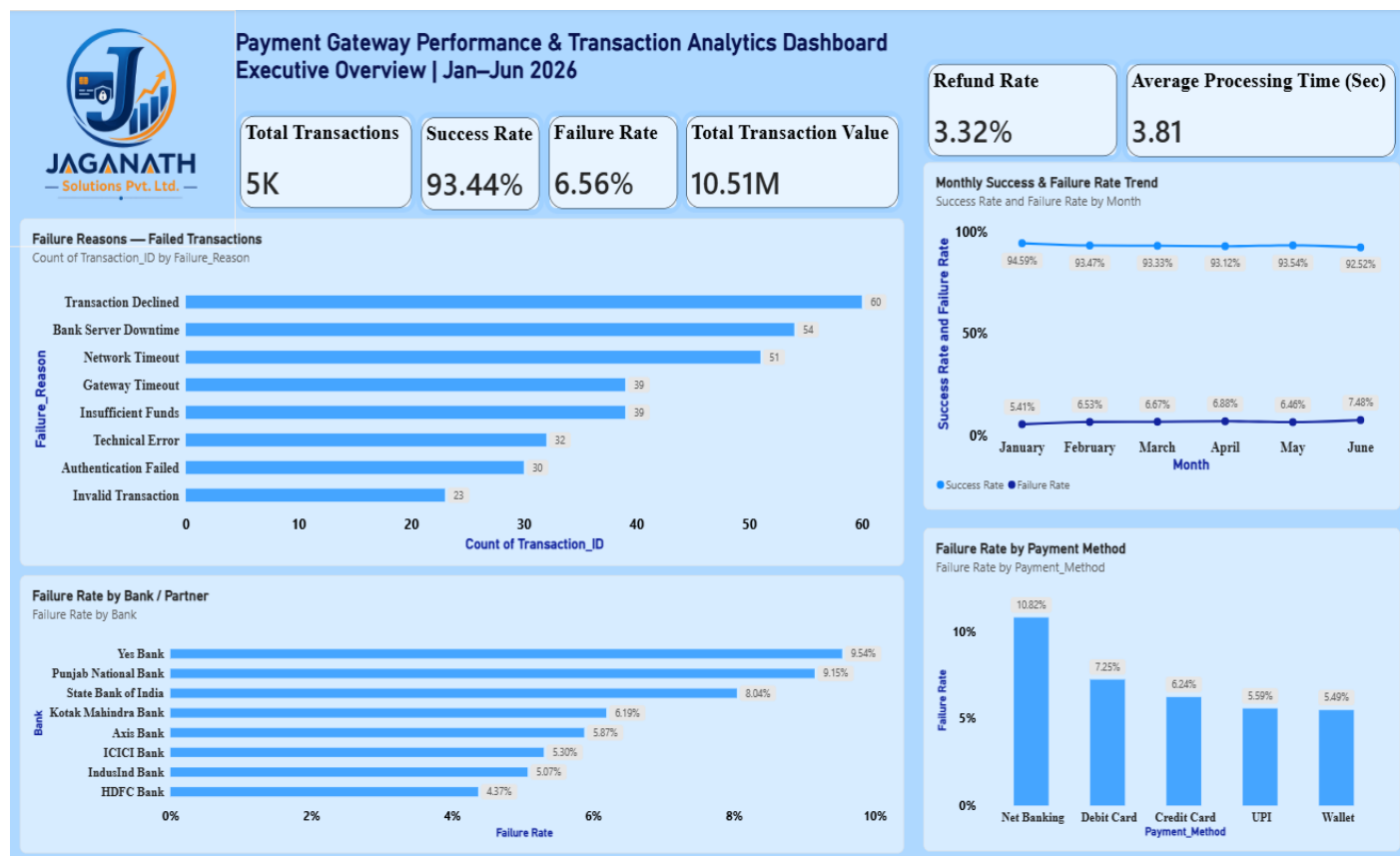
Dashboard User Flow

1. Start with the KPI cards. → 2. Check the trend. → 3. Identify the top failure reasons. → 4. Compare payment methods. → 5. Drill into banks/merchants/regions. → 6. Prioritize an improvement action.

Important BA Point

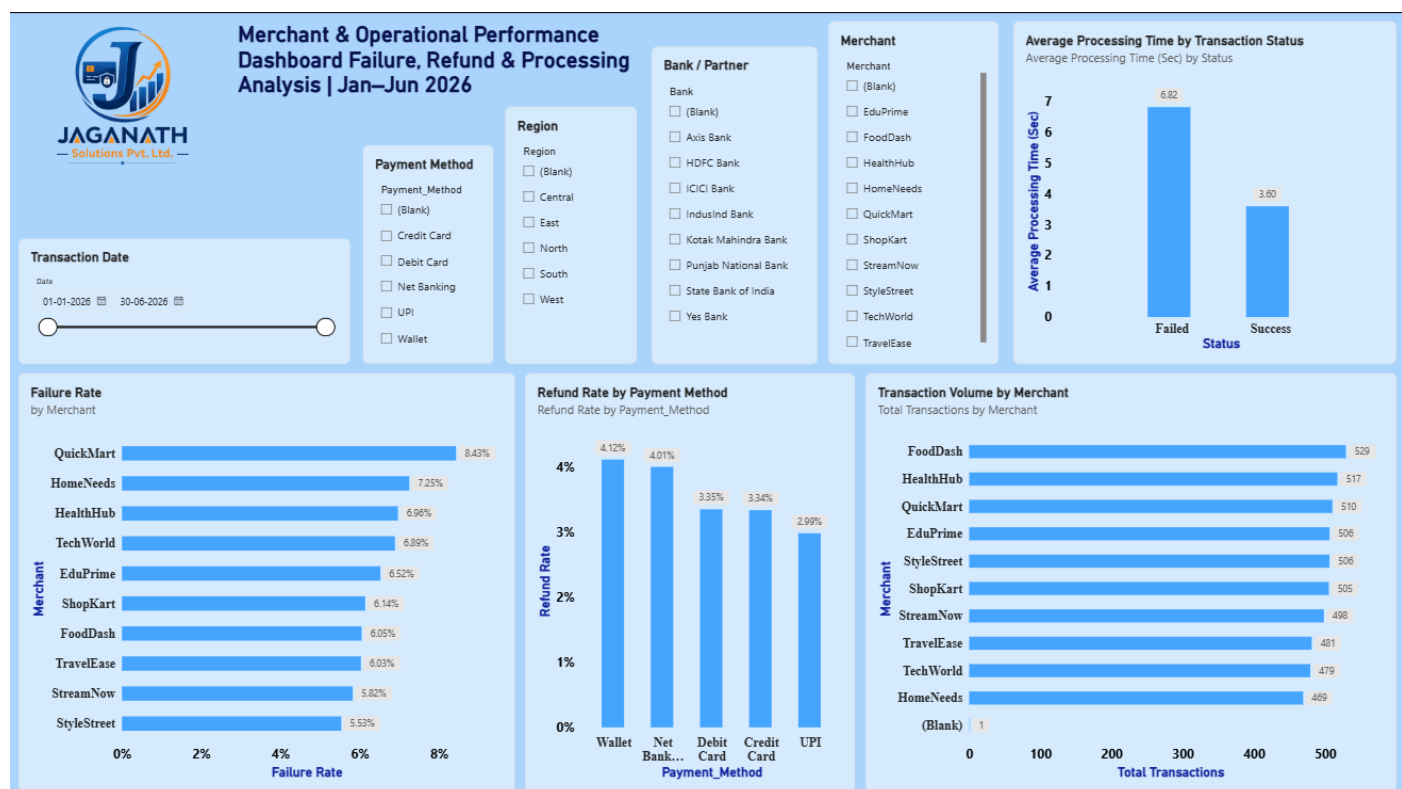
The dashboard is not the final solution by itself. Its purpose is to provide reliable visibility that helps Product, Operations, Finance, Engineering and Support teams make evidence-based decisions.

Power BI Dashboard — Page 1: Executive Overview



Purpose: Provides an executive view of transaction volume, payment success/failure performance, failure reasons, banking-partner performance and payment-method performance.

Power BI Dashboard — Page 2: Merchant & Operational Analysis



Purpose: Provides detailed analysis of merchant failure rates, transaction volume, processing time, refund rates and interactive filtering by date, payment method, bank, merchant and region.

7. Actual Business Insights

Insight 1 — Failure Concentration: The analysis identified 328 failed transactions out of 5,000 total transactions. The three leading failure reasons were Transaction Declined (60), Bank Server Downtime (54), and Network Timeout (51). Together, these three categories account for 165 failed transactions, representing approximately 50.3% of all failures.

BA Interpretation: The concentration of failures in a small number of categories provides a clear starting point for operational and technical investigation.

Insight 2 — Payment Method Performance: Net Banking recorded the highest failure rate at 10.82%, compared with 7.25% for Debit Card, 6.24% for Credit Card, 5.59% for UPI and 5.49% for Wallet.

BA Interpretation: Net Banking represents a key area for further investigation because its observed failure rate is higher than the other payment methods in the dataset.

Insight 3 — Monthly Failure Trend: The overall failure rate increased from 5.41% in January to 7.48% in June, with June showing the highest observed monthly failure rate during the January–June 2026 analysis period.

BA Interpretation: The movement in failure rate across the period indicates that time-based patterns should be examined alongside payment-method, bank and failure-reason analysis.

Insight 4 — Merchant Performance:

- QuickMart → 8.43%
- HomeNeeds → 7.25%
- HealthHub → 6.96%
- TechWorld → 6.89%

QuickMart recorded the highest observed merchant-level failure rate at 8.43%, followed by HomeNeeds at 7.25%. This indicates variation in payment reliability across merchants and supports further investigation of merchant-specific transaction patterns.

Insight 5 — Processing Time:

Failed = 6.82 sec

Success = 3.60 sec

Failed transactions had an average processing time of 6.82 seconds compared with 3.60 seconds for successful transactions. This indicates a noticeable difference in processing duration between successful and failed outcomes and warrants further investigation into timeout and processing-related failure patterns.

8. Root-Cause Analysis

Observed Pattern	Evidence	Investigation Area
Failure concentration	Top 3 reasons account for 50.3% of failures	Failure reason investigation
Higher Net Banking failure rate	10.82%	Payment method and bank analysis
Partner-level variation	Bank failure rates vary across partners	Bank/partner investigation
Higher processing time for failures	6.82 sec vs 3.60 sec	Timeout and processing investigation
Merchant-level variation	QuickMart 8.43% vs other merchants	Merchant-level investigation

The analysis identifies these areas as evidence-based investigation points. Root causes should be confirmed through transaction-level analysis and supporting evidence rather than inferred from correlation alone.

9. Evidence-Based Recommendations

Recommendation 1 — Failure Reason Monitoring

Prioritize monitoring and investigation of the three highest-volume failure categories: Transaction Declined, Bank Server Downtime and Network Timeout, which together account for approximately 50.3% of observed failures.

Expected Benefit: Focus operational and technical investigation on the largest observed failure drivers.

Recommendation 2 — Net Banking Investigation

Conduct a focused review of Net Banking transactions and associated banking partners because Net Banking recorded the highest payment-method failure rate at 10.82%.

Expected Benefit: Identify payment-method or partner-specific reliability patterns.

Recommendation 3 — Processing-Time Investigation

Investigate transactions with extended processing times, particularly failed transactions, which averaged 6.82 seconds compared with 3.60 seconds for successful transactions.

Expected Benefit: Identify potential timeout or processing-delay patterns contributing to failed outcomes.

Recommendation 4 — Merchant Monitoring

Establish merchant-level monitoring for higher observed failure rates, with QuickMart recording the highest merchant-level failure rate at 8.43% in the analyzed dataset.

Expected Benefit: Enable targeted investigation of merchant-specific transaction patterns.

10. Business Impact + Conclusion

Expected Business Impact

Business Area	Potential Impact
Revenue	Lower revenue leakage from unsuccessful transactions
Customers	Fewer payment drop-offs and retries
Merchants	Improved payment conversion and trust
Operations	Lower investigation and reconciliation workload
Support	Fewer payment-failure complaints
Product / Engineering	Clearer prioritization of reliability improvements
Management	Better data-driven visibility and decision-making

Success Criteria

- Major payment failure drivers identified with analytical evidence.
- Business impact quantified where data permits.
- BRD and FRD requirements documented.
- AS-IS and TO-BE processes completed.
- Excel/Power BI analysis completed as applicable.
- Relevant payment-performance KPIs established.
- Recommendations linked directly to findings.
- Expected solution benefits documented.
- Actual business insights and recommendations linked to validated dashboard findings.

Conclusion

This project demonstrates how a Business Analyst can approach a real-world-style payment reliability problem without using a complex framework. The BA starts with the business problem, understands stakeholders and the current process, defines requirements, structures the data questions, creates simple KPIs, uses Power BI for visibility, identifies failure patterns, and translates findings into actionable recommendations.

For an entry-level BA portfolio, the strength of this project is not the size of the document but the clarity of the problem-to-requirement-to-analysis-to-recommendation journey.